Detailed Evaluation of Ecosystem Processes (DEEP) Field Worksheet

Northeastern Highlands reference assessment | version 10
USEPA NRSA 2013-24 (L3 ecoregion 58), StreamCurves regional analysis, methodology 0.16-provisional

Reach ID: COMID 9327042 **Reach Length:** 1,234 ft **Date:** _____ **Assessor(s):** _____ **Coordinates:** 43.70000, -72.20000
Site: (unnamed reach) | drainage area 3.0 km2 | NHDPlus V2 basin of the nearest StreamCat reach (StreamCat lookup engine)

RECORDING INSTRUCTIONS

Record the measured value of each metric in its units. Metrics measured in the field are marked F and metrics answered from the desk are marked D. A value printed in the Value cell was already answered from the desk and carries its source. Enter the values in DEEP, which reads each one on the metric's reference curve to give an index from 0 to 1. The function score is the mean index of the function's metrics times 15 and reads Functioning (15 to 11), Functioning At-Risk (10 to 6), or Non-Functioning (5 to 0). A metric that does not apply is marked NA and drops out of the score. Where a metric lists curve sets, mark the set that fits the reach.

Function	Metrics	Value	Index
HYDROLOGY FUNCTIONS			
Catchment hydrology: Runoff and infiltration sustain natural flow regime, carry appropriate sediment and nutrients from uplands, and reliably cue spawning/migration of aquatic life.	**Impervious surface** (%) D: Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. *DESKTOP: NLCD 2021*	**3.0**	
	Agriculture, crop and hay (%) D: Percent of the delineated watershed in cultivated crops plus hay and pasture, computed by DEEP. No field measurement. *DESKTOP: EPA StreamCat*	**12.75**	
Score:	**Notes/Other Metrics:**		
Surface water storage: Wetlands and storage features store floodwater, recharge groundwater, sustain baseflow, and provide low-velocity habitat.	**Wetland** (%) D: Percent of the watershed in woody plus herbaceous wetland (NLCD 2019), read from EPA StreamCat for the reach. No field measurement. *DESKTOP: Computed by DEEP (reference only)*	**0.4**	
Score:	**Notes/Other Metrics:**		
Reach inflow: Quantity + quality of inflow (tributaries, ditches, and pipes) does not provide harmful peaks or pollution, and supports diverse habitat conditions.	**Road density** (km/sq km) D: Road length per square kilometer of the delineated watershed, computed by DEEP. No field measurement. *DESKTOP: NLCD 2021*	**3.0**	
Score:	**Notes/Other Metrics:**		
Streamflow regime: Integrates the range of "typical" flows experienced by other processes. Examines the degree to which upstream infrastructure or land uses have fundamentally altered flow regimes (e.g., hydropeaking, dams, withdrawals).	**Base flow index** (%) D: Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. *DESKTOP: EPA StreamCat*	**12.75**	
	Degree of regulation (%) D: Normal storage of upstream dams as a percent of mean annual runoff, both read from EPA StreamCat for the reach. No field measurement. *DESKTOP: Computed by DEEP*	**0.4**	
Score:	**Notes/Other Metrics:**		

HYDRAULICS FUNCTIONS			
Low flow and baseflow dynamics: Ensures habitat availability and water quality during low water levels, and indicates non-storm conditions experienced the majority of the time.	**Base flow index** (%) D: Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. *DESKTOP: EPA StreamCat*	**12.75**	
	Bankfull width:depth ratio F: At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
	Residual pool depth not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
High flow dynamics: Peak flows reshape channel without chronic instability, redistribute wood and sediment, reset habitats, and supply nutrients to floodplains.	**Impervious surface** (%) D: Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. *DESKTOP: NLCD 2021*	**3.0**	
Score:	**Notes/Other Metrics:**		
Floodplain connectivity: Frequent overbank flows reduce flood peaks, support riparian vegetation, create off-channel refugia, and extend nutrient processing time.	**Bankfull width:depth ratio** F: At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
	Bankfull height above channel (m) F: At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights. Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
Score:	**Notes/Other Metrics:**		
Hyporheic connectivity: Surface water-groundwater exchange moderates temperature, transforms nutrients, supplies oxygen, and supports aquatic life.	**Sand + fines** (%) F: Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer.		
	Embeddedness (%) F: At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates.		
Score:	**Notes/Other Metrics:**		
GEOMORPHOLOGY FUNCTIONS			
Channel evolution: Long-term changes in channel size and slope maintain dynamic equilibrium and diverse habitats while interacting with the floodplain.	**Bankfull width:depth ratio** F: At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
	Bankfull height above channel (m) F: At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights. Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
Score:	**Notes/Other Metrics:**		

Channel and floodplain dynamics: Examines the role of bank processes and erosion in channel change. Observes patterns in sinuosity and curvature affecting habitat complexity.	**Sinuosity** F: Take a compass backsight bearing and the distance between each pair of neighboring transects. Value: channel length along those segments divided by the straight-line distance between the reach ends.		
	Bank angle (degrees) F: On both banks of each transect, lay a pole about 1 m long against the bank and read its angle from horizontal with a clinometer, in degrees. Value: mean of the 22 readings.		
Score:	**Notes/Other Metrics:**		
Sediment continuity: Balanced sediment supply and transport preserves bed elevations, substrate sizes, spawning/benthic habitats, and supports riparian succession.	**Relative bed stability** (log) F: Computed: the base 10 log of the geometric mean particle diameter divided by the largest diameter the bankfull flow can move, from the particle count, reach slope, bankfull depth, wood tally and residual pools (NRSA physical habitat calculations). Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
	Sand + fines (%) F: Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer.		
Score:	**Notes/Other Metrics:**		
Bed composition and bedform dynamics: Supports aquatic habitats through streambed material and bedform dynamics. Enhances habitat complexity and streambank stability.	**Substrate size** (log mm) F: From the 105-particle size class count (as for sand and fines). Value: base 10 log of the geometric mean particle diameter in mm, from the size classes (NRSA physical habitat calculations).		
	Large wood volume (m3/100m) F: Between each pair of transects, tally every piece of wood at least 10 cm in diameter and 1.5 m long in or above the bankfull channel, by length and diameter class. Value: wood volume per 100 m of channel, from the tally.		
Score:	**Notes/Other Metrics:**		
PHYSICOCHEMISTRY FUNCTIONS			
Light & thermal regime: Shade, solar input, and groundwater keep temperature and light within natural ranges for stream metabolism and dissolved oxygen.	**Chlorophyll a** (ug/L) F: Water-column chlorophyll a (ug/L) from a 2 L mid-channel grab sample at the X-site, collected in an amber bottle, filtered and analyzed in a lab.		
	Canopy density, mid-channel (%) F: At mid-channel on each transect, read a convex spherical densiometer facing upstream, downstream, left and right (covered points of 17). Value: mean of the 44 readings, as a percent. Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
Score:	**Notes/Other Metrics:**		
Carbon processing: Organic matter is captured and broken down, fueling food webs, balancing production/ respiration, moderating pH/ redox, and supplying nutrients.	**Riparian veg cover, woody** (frac) F: In a 10 m by 10 m plot on each bank at each transect, estimate the areal cover class of woody vegetation in the canopy, understory and ground layer. Value: summed woody cover of the 3 layers, averaged over the 22 plots, as a fraction.		
Score:	**Notes/Other Metrics:**		
Nutrient cycling: Nitrogen and phosphorus transformations feed primary production, regulate excess loads, and store or release nutrients.	**Total nitrogen** (mg N/L) F: Total nitrogen (mg N/L), lab analysis of the unfiltered mid-channel grab sample from the X-site, kept on ice.		
	Total phosphorus (ug/L) F: Total phosphorus (ug/L), lab analysis of the mid-channel grab sample from the X-site, kept on ice.		
Score:	**Notes/Other Metrics:**		

Water & soil quality: Physical and chemical processes limit pollutants, retain beneficial constituents, and protect ecological and human uses.	**Specific conductivity** (uS/cm) F: Specific conductance (uS/cm). NRSA reports the lab value of a mid-channel grab sample at the X-site (the reach center); a calibrated multi-parameter meter reads the same quantity in the stream there.		
	Turbidity (NTU) F: Turbidity (NTU), lab analysis of the mid-channel grab sample from the X-site. Sample before anyone walks the channel upstream.		
Score:	**Notes/Other Metrics:**		
BIOLOGY FUNCTIONS			
Habitat provision: Channel and floodplain structure supply depth, velocity, substrate diversity, and vegetation to support native organisms through all life stages.	**Fast-water / riffle habitat** (%) F: At each thalweg station, record the channel unit (falls, cascade, rapid, riffle, glide, pool, dry). Value: percent of stations in falls, cascade, rapid or riffle. Curve set: [ ] All streams (pooled) [ ] Moderate gradient (0.5 to 2 percent) [ ] Steep (2 percent and above)		
	Natural fish cover (frac) F: In the 10 m of channel centered on each transect, estimate the areal cover class of each fish cover type on the form (wood, brush, overhanging vegetation, undercut banks, boulders, others). Value: summed cover of the natural types, averaged over the transects. Curve set: [ ] All streams (pooled) [ ] Headwater (10 km2 and under) [ ] Small stream (10 to 100 km2)		
Score:	**Notes/Other Metrics:**		
Population support: Physical and chemical conditions enable spawning, juvenile growth, migration, and stress survival of target taxa.	**Benthic total taxa richness** F: From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: total number of taxa.		
	Native fish taxa richness not scored *Insufficient reference support. No value is needed.*	**NA**	
Score:	**Notes/Other Metrics:**		
Community dynamics: Species interactions sustain biodiversity, restrain invasives, and build resilience to disturbance.	**Shannon diversity (H')** (H') F: From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: Shannon diversity of the taxa counted.		
	Tolerant individuals (%) ~ HBI F: From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: percent of individuals in pollution-tolerant taxa.		
Score:	**Notes/Other Metrics:**		
Watershed connectivity: Continuous pathways for free movement of water, sediment, and organisms supporting recolonization, maintaining species diversity, and enabling recovery.	**Mapped dams within one mile** (dams) D: Dams in the USACE National Inventory of Dams within 1 mile of the site, counted by DEEP. Note in the field any barrier the inventory does not show. *DESKTOP: NLCD 2021*	**3.0**	
Score:	**Notes/Other Metrics:**		

METRICS NOT SCORED

Native fish taxa richness, Residual pool depth. These metrics were considered for this assessment. These metrics are not scored. A metric withheld for insufficient reference support has no curve. No pool of least-disturbed stations in this ecoregion or its parents supports it, no national donor pool was shown to transfer, no modeled expectation passed validation, and no published criterion applies. No measurement is needed for the score. A value may still be recorded under Notes.